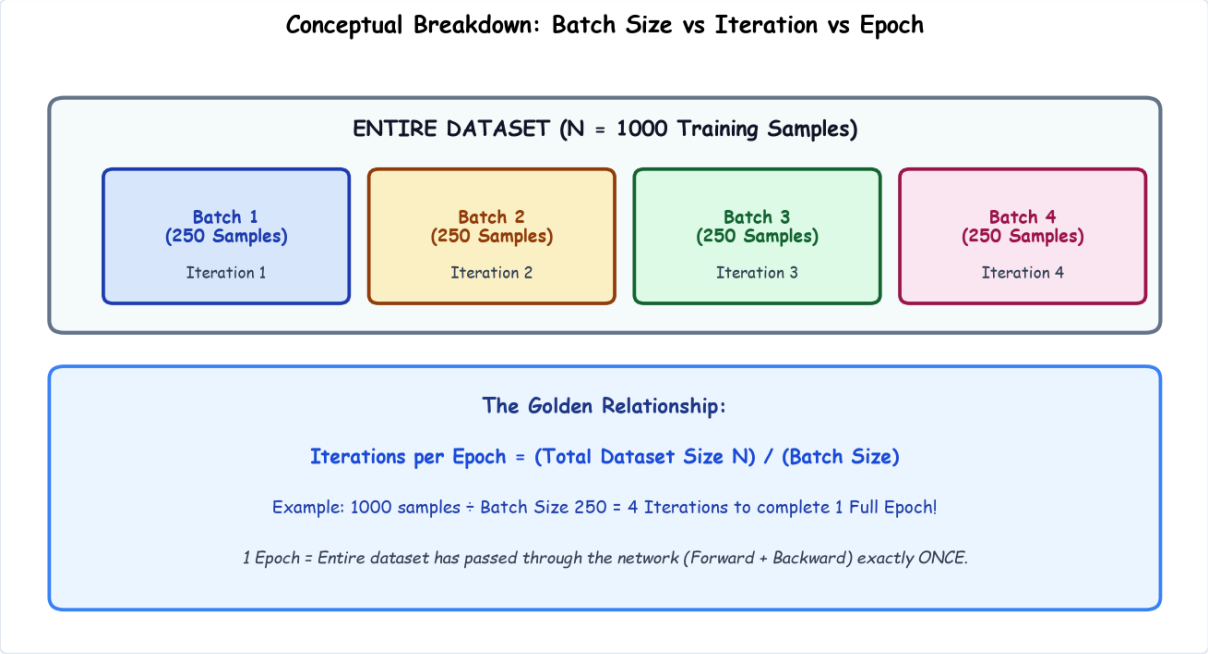
Submodule 37.08: Batch vs Iteration vs Epoch

1. Introduction: Deconstructing Training Terminology

When configuring neural network training pipelines, three foundational hyperparameters govern data ingestion, memory utilization, and optimization cadence: **Batch**, **Iteration**, and **Epoch**. Confusing these metrics leads to incorrect learning rate scaling and suboptimal computational resource allocation.



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2. Detailed Technical Definitions

1. Batch & Batch Size (B)

- **Batch**: A discrete subset of training examples passed simultaneously into the neural network for forward propagation, loss evaluation, and backward gradient computation.
- **Batch Size (B)**: The integer number of individual data samples contained within one batch.
    - Standard deep learning batch sizes are powers of two (e.g., $16, 32, 64, 128, 256$) to optimize parallel tensor alignment in GPU warp/thread block architectures.

2. Iteration (Step)

- An **Iteration** (often termed a training *Step*) represents **one complete cycle of Forward Pass + Backward Pass + Weight Update on exactly ONE batch**.
- If a dataset is partitioned into K batches, completing all batches requires K iterations.

3. Epoch

- An **Epoch** represents **one complete, exhaustive pass of the ENTIRE training dataset** through the neural network.
- During 1 epoch, every single training sample in the dataset has been seen by the model exactly once.

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3. The Governing Mathematical Relationship

Given:

- Total number of training samples in dataset: N
- Configured Batch Size: B

The number of iterations (weight updates) required to complete **one single epoch** is:

$$\text{Iterations per Epoch} = \left\lceil \frac{N}{B} \right\rceil$$

Concrete Numerical Walkthrough

Suppose an industrial computer vision dataset contains $N = 50,000$ images:

- **Scenario A: Batch Size** $B = 100$

$$\text{Iterations per Epoch} = \frac{50,000}{100} = 500 \text{ weight updates per epoch}$$

If trained for $E = 20$ epochs:

$$\text{Total Training Iterations} = 500 \times 20 = 10,000 \text{ steps}$$

- **Scenario B: Batch Size** $B = 64$

$$\text{Iterations per Epoch} = \left\lceil \frac{50,000}{64} \right\rceil = 782 \text{ weight updates per epoc}$$

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4. Architectural Trade-offs of Batch Size Selection

Batch Size Setting	GPU Memory Footprint	Gradient Estimation Quality	Training Wall-Clock Speed	Generalization Ability
Small (e.g., $8 - 32$)	Minimal VRAM	High stochastic noise	Slower (underutilizes GPU parallelism)	Superior (stochastic noise escapes sharp local minima)
Moderate (e.g., $64 - 128$)	Optimal balance	Balanced variance	Fast (efficient GPU tensor cores utilization)	Strong generalization

| **Large (e.g., $1024 - 4096$)** | Massive VRAM | Low variance (approaches true gradient) | Extremely fast throughput | Risks converging to sharp minima (poorer test generalization) |\n